INVESTIGATING THE DIFFERENCE BETWEEN TWO ENDOLYSINS ENCODED BY PHAGES IN CLUSTER BF

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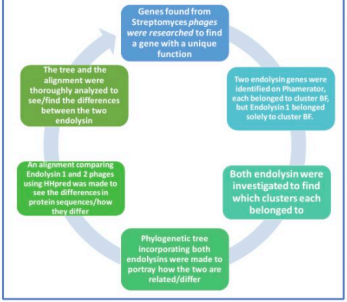
Background

- Endolysins are enzymes that are used in the lytic cycle to break down a bacterial cell wall from the inside when activated.
- Bacteriophage endolysins can be used to treat bacterial infections by acting as antibacterial agents.
- Two types of endolysin were found in cluster BF. One endolysin was found to only cluster BF (Endolysin 1) while another one was found in multiple clusters (Endolysin 2).

Research Question

How is Endolysin 1 present in the cluster BF while Endolysin 2 is present in clusters BF, BD, and BC?

Methods



Results

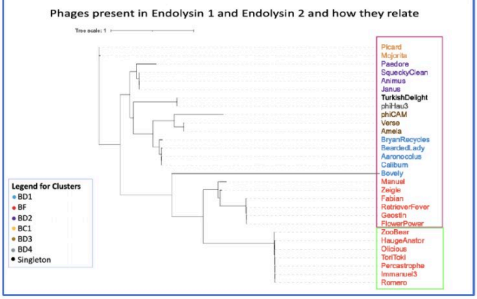


Figure 1. Phylogenic tree of endolysin protein sequences using VICTOR. The phylogenic tree above shows all the phages that contain Endolysin 1 and Endolysin 2. The pink box surrounds the phages with Endolysin 2 and the green box surrounds phages with Endolysin 1. It is depicted that cluster BF phages from both Endolysin 1 and 2 are closely related. The tree was generated using the program VICTOR.

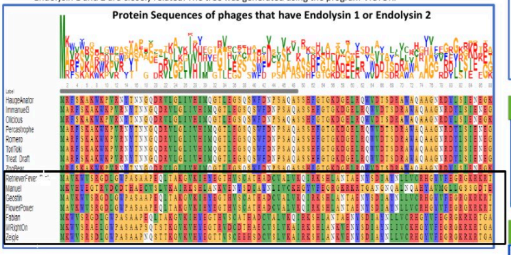


Figure 2: The graph above generated using HHpred represents the similarities in protein sequences of phages that have Endolysin 1 or Endolysin 2 in cluster BF. There is great similarity in protein sequence of phages that have Endolysin 1. However, there is little similarity in protein sequence of phages that have Endolysin 2. Romero and ToriToki contain Endolysin 1 and are highly similar , but ToriToki and Manuel that contain Endolysin 2 have little similarity. Some protein sequences are conserved by Endolysin 1 and Endolysin 2 phages. This shows how the two endolysins are different but related .

Discussion

- It is evident that Endolysin 1 and Endolysin 2 are related with common ancestors.
- Endolysin 2 is older compared to Endolysin 1 because Endolysin 2 is found in more clusters and phages than Endolysin 1.
- Endolysin 1 is not conserved however, there were locations conserved by both
- Some protein sequences of phages code for different proteins with very similar functions which indicates that the two endolysins came from the same ancestor but over time evolved into different types.

Future Directions

- Investigate why Endolysin 1 only appears in cluster BF while Endolysin 2 appears in multiple clusters by doing a protein expression analysis. The endolysin protein can be expressed as a recombinant protein and inserted into different hosts to do further analysis.

References/Citations

"The Actinobacteriophage Database: Home." *The Actinobacteriophage Database | Home*.

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"VICTOR Virus Classification and Tree Building Online Resource." *Ggdc.dsmz.de*, ggdc.dsmz.de/victor.php.

Acknowledgement

This research was supported by the Science Education Alliance, Howard Hughes Medical Institute, Chevy Chase, MD, and the National Institute of General Medical Sciences of the National Institutes of Health under Award Numbers TL4GM118989, UL1GM118988, and RL5GM118987. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health.